Chap 5. Stochastic integration Itô's formula

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Outline

Itô's integration

Main focus.

Recall from Section 3 that Brownian motion B can be enhanced to a rough path $\mathrm{B} = (B, \mathbb{B})$.

$\mathbb{B}_{s,t} = \mathbb{B}^{It\hat{o}}_{s,t} := \int_s^t B_{s,u} \otimes dB_u$ and enhanced Brownian motion has almost surely(non-geometric) α-Hölder rough sample paths, for any $\alpha \in (\frac{1}{3}, \frac{1}{2})$

► Rough integrals against($\mathrm{B} = \mathrm{B}^{It\hat{o}}$) = Itô's integrals whenever both are defined.

Itô's integration

Proposition 5.1

Assume $(Y(\omega), Y'(\omega)) \in \mathscr{D}^{2\alpha}_{B(W)}$ for every $\omega \in N_2^c$.
Let $\mathrm{B}(\omega) = (B(\omega), \mathbb{B}(\omega)) \in \mathscr{C}^{\alpha}$ for every $\omega \in N_1^c$. Set $N_3 = N_1 \cup \ N_2$.
Then the rough integral

$$\int_0^T Y dB = \lim_{n \to \infty} \sum_{[u,v] \in \mathcal{P}_n} (Y_u B_{u,v} + Y'_u \mathbb{B}_{u,v}) \tag{1}$$

exists, for each fixed $\omega \in N_3^c$, along any sequence $(\mathcal{P}_n)$ with mesh $|\mathcal{P}_n| \downarrow 0$. If Y,Y' are adpated then, almost surely, $\int_0^T Y d\mathrm{B} = \int_0^T Y dB$.

Itô's integration

Proof.

WLOG let $T = 1$.

From Gubinelli's Thm, the existence of the rough integral for $\omega \in N_3^c$ under the stated condition is immediate, applied to Y(w), controlled by $B(\omega)$, for $\omega \in N_2^c$.

Recall that for any continuous, adpated process Y the Itô integral against Brownian motion has the representation:

$$\int_0^1 YdB = \lim_{n\to\infty} \sum_{[u,v]\in\mathcal{P}_n} Y_u B_{u,v} \tag{2}$$

(in probability) along any sequence $(\mathcal{P}_n)$ with mesh $|\mathcal{P}_n| \downarrow 0$.
By switching to a subsequence, if necessary, we can assume that the convergence holds almost surely, say on N_4^c. Set $N_5 := N_3 \cup \ N_4$.

Itô's integration

Thus, we suffice to show to show that following claim.

Claim

Rough and Itô integral coincide on N_5^c

proof of claim : With a look at the respective Riemann-sums, convergent away from N_5, basic analysis tell us that

$$\forall \omega \in N_5^c : \exists \lim_n \sum_{[u,v] \in \mathcal{P}_n} Y'_u \mathbb{B}_{u,v}, \tag{3}$$

and that the limit eqauls the difference of rough and Itô integrals. Of course, $|\mathcal{P}_n| \downarrow 0$ and to see that the above limit is zero, we only need to show that

$$\mathcal{O}(|\mathcal{P}|) = \| \sum_{[u,v] \in \mathcal{P}_n} Y'_u \mathbb{B}_{u,v} \|_{L^2}^2 \tag{4}$$

Itô's integration

To complete this,
assume the partition is of the form $\mathcal{P} = 0 = \mathcal{T}_0 < \mathcal{T}_1 < \cdots < \mathcal{T}_N = 1$,
and define a (discrete time) martingale as follows:

$$\mathcal{S}_0 := 0, \quad \mathcal{S}_{k+1} - \mathcal{S}_k = Y'_{\mathcal{T}_k} \mathbb{B}_{\mathcal{T}_k, \mathcal{T}_{k+1}}. \tag{5}$$

since $|\mathbb{B}_{\mathcal{T}_k, \mathcal{T}_{k+1}}|^2_{L^2}$ is proportional to $|\mathcal{T}_{k+1} - \mathcal{T}_k|^2$, therefore, we have :

$$\begin{aligned} \| \sum_{[u,v] \in \mathcal{P}_n} Y'_u \mathbb{B}_{u,v} \|^2_{L^2} = | \sum_{k=0}^{N-1} (\mathcal{S}_{k+1} - \mathcal{S}_k)|^2_{L^2} &= \sum_{k=0}^{N-1} |\mathcal{S}_{k+1} - \mathcal{S}_k|^2_{L^2} \\ &\leq M^2 \sum_{k=0}^{N-1} |\mathbb{B}_{\mathcal{T}_k, \mathcal{T}_{k+1}}|^2_{L^2} = \mathcal{O}(|\mathcal{P}|) \end{aligned}$$

We complete the proof of bounded case ($\sup_{\omega \in \mathbb{N}_5^c} |Y'(\omega)|_\infty \leq M$).

Itô's integration

Let's see the unbounded case. It suffice to show the convergence of the following in probability,

$$\mathcal{O}(|\mathcal{P}|) = \| \sum_{[u,v]\in\mathcal{P}_n} Y'_u \mathbb{B}_{u,v} \|_{L^2}^2 \tag{6}$$

To complete this case, we introduce stopping times:

$$\mathcal{T}_M := max\{t \in \mathcal{P} : |{Y_t}'| < M|\} \in [0, T] \cup \ \{\infty\} \tag{7}$$

Note that $\mathcal{T}_M = \infty$ as M $\rightarrow \infty$. The stopped process $S^{\mathcal{T}_M}$ is also a martingale, and we see as above that, for every fixed $M > 0$,

$$\mathcal{O}(|\mathcal{P}|) = \| \sum_{\substack{[u,v]\in\mathcal{P}_n \\ u\leq\mathcal{T}_M}} Y'_u \mathbb{B}_{u,v} \|_{L^2}^2 \tag{8}$$

If we send M to ∞, we are done.

Stratonovich integration

Stratonovich integration

Main recall 1.

$\mathbb{B}^{Strat}_{s,t} := \int_s^t B_{s,u} \otimes \circ dB_u = \mathbb{B}^{It\hat{o}}_{s,t} + \frac{1}{2}(t-s)\mathrm{I}$

- A.s, geometric α-Hölder rough sample paths for any $\alpha \in (\frac{1}{3}, \frac{1}{2})$

Main recall 2. (Stratonovich integral)

$$\int_0^T Y \circ dB := \int_0^T YdB + \frac{1}{2}[Y, B]_T \tag{9}$$

whenever the Itô's integral is well defined and the quadratic covariation of Y and B exists in the sense that

$$[Y, B]_T := \lim_{|\mathcal{P}| \to 0} \sum_{[u,v] \in \mathcal{P}} Y_{u,v} B_{u,v}. \tag{10}$$

in probability.

Stratonovich integration

Main focus.

What are natural assumptions guranteeing that both notions of integral are well defined for which folloiwing eqaulity holds?

Rough intergration against Stratonovich enhanced Brownian motion
$=$ Usual Stratonovich integration against Brownian motion

Stratonovich integration

Lemma 5.2

Let $\mathbf{B}(\omega) = (B(\omega), \mathbb{B}(\omega)) \in \mathscr{C}^\alpha$ for every $\omega \in N_1^c$.
Assume Y=Y(ω) $\in \mathcal{C}^\alpha_{B(\omega)}$ for every $\omega \in N_2^c$. Set $N_3 = N_1 \cup \ N_2$. Then the rough integral

$$\int_0^T YdB = \lim_{n\to\infty} \sum_{[u,v]\in\mathcal{P}_n} (Y_u B_{u,v} + Y_u' \mathbb{B}^{Strat}_{u,v}) \tag{11}$$

exists, Moreover if Y,Y' are adpated then, the quadratic covariation of Y and B exists and, almost surely, $\int_0^T Yd\mathbf{B} = \int_0^T Y \circ \mathrm{d}B$.

Itô's formula and Föllmer

Main focus

Given a smooth path $X : [0, T] \to V$ and a map $F : V \to W$ in $\mathcal{C}_b^1$, where V,W are Banach space, the chain rule from classical calculus:

$$F(X_t) = F(X_0) + \int_0^t DF(X_s) X_s, \quad 0 \leq t \leq T \tag{12}$$

Our main focus is the following

Question

Does a same change of variables formula holds for geometric rough paths $X = (\mathrm{X}, \mathbb{X})$?
"Second order" correction involving $D^2 F$ appears in non geometric sense?

Main preliminary

From chapter 1 to 4 , we were devoted to understand rough integration against 1-forms, say G=G(X) and we found:

$$\int G(X)dX \approx \sum_{[s,t]\in\mathcal{P}} \langle G\left(X_s\right), X_{s,t}\rangle + \langle DG\left(X_s\right), \mathbb{X}_{s,t}\rangle \tag{13}$$

Let split $\mathbb{X}$ into symmetric part and antisymmetric part.

$$\mathbb{S}_{s,t} := Sym(\mathbb{X}_{s,t}), \quad Anti(\mathbb{X}_{s,t}) := \mathbb{A}_{s,t} \tag{14}$$

Then

$$\langle DG\left(X_s\right), \mathbb{X}_{s,t}\rangle = \langle DG\left(X_s\right), \mathbb{S}_{s,t}\rangle + \langle DG\left(X_s\right), \mathbb{A}_{s,t}\rangle \tag{15}$$

Itô's formula and Föllmer

- Observe that the final term of (15) disappears when G=DF.
- In other words, antisymmetric is key point for general integrals of 1-form.

Definition5.3

We call $\mathbf{X} = (X,\mathbb{S})$ a reduced rough path, in symbols $\mathbf{X} \in \mathscr{C}_{\Gamma}^{\alpha}([0,\mathcal{T}],V)$ if $X{=}X_t$ takes values in a Banach space V, $\mathbb{S}{=}\mathbb{S}_{s,t}$ takes values in $\mathrm{Sym}(V \otimes V)$, and the following hold:

i) a "reduced" Chen relation

$$\mathbb{S}_{s,t} - \mathbb{S}_{s,u} - \mathbb{S}_{u,t} = Sym(X_{s,u} \otimes X_{u,t}), \quad 0 \leq s,t,u \leq \mathcal{T} \tag{16}$$

ii) $\mathbb{X}_{s,t} = \mathcal{O}(|t-s|^{\alpha}), \quad \mathbb{S}_{s,t} = \mathcal{O}(|t-s|^{2\alpha})$, for some $\alpha > \frac{1}{3}$.

Lemma 5.4

Given $X \in \mathcal{C}^\alpha$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, the "geometric" choice $\bar{\mathbb{S}}_{s,t} = \frac{1}{2}\mathbb{X}_{s,t} \otimes \mathbb{X}_{s,t}$ yields a reduced rough path, i.e. $(X, \bar{\mathbb{S}}) \in \mathscr{C}_\Gamma^\alpha$.
Moreover, for any 2α-Hölder path γ with values in $\mathrm{Sym}(V \otimes V)$, the perturbation

$$\mathbb{S}_{s,t} = \bar{\mathbb{S}}_{s,t} + \frac{1}{2}(\gamma_t - \gamma_s) = \frac{1}{2}(X_{s,t} \otimes X_{s,t} + \gamma_{s,t}) \tag{17}$$

also yields a reduced path $(X, \mathbb{S})$. Finally, all reduced rough path over X are obtained in this fashion.

- This Lemma gives a one to one correspondence between $\mathbb{S}$ and γ.
- We need to observe the role of γ

Definition5.5(Bracket of a reduced rough path)

Given $\mathbf{X} = (X, \mathbb{S}) \in \mathscr{C}_{\Gamma}^{\alpha}(\mathrm{V})$, we define the bracket

$$\begin{aligned} [\mathbf{X}] : [0, T] &\to Sym(V \otimes V) \\ t &\mapsto [\mathbf{X}]_t := X_{0,t} \otimes X_{0,t} - 2\mathbb{S}_{0,t}. \end{aligned}$$

- Note that $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$ by previous Lemma.
- If $[\mathbf{X}]_{s,t} := \mathrm{X}_{s,t} \otimes \mathrm{X}_{s,t} - 2\mathbb{S}_{s,t}$, then $[\mathbf{X}]_{s,t} = [\mathbf{X}]_{0,t} - [\mathbf{X}]_{0,s}$ for any two times s,t.

Proposition 5.6(Itô's formula for reduced rough path)

Let $F : V \to W$ be of $\mathcal{C}_b^3$ and let $\mathbf{X}$=(X,$\mathbb{S}$)$\in \mathscr{C}_\Gamma^\alpha([0,\mathcal{T}],V)$ with $\alpha > \frac{1}{3}$. Then

$$F(X_t) = F(X_0) + \int_0^t DF(X_s)\, d\mathbf{X}_s + \int_0^t D^2F(X_s)d[X]_s, \quad 0 \le t \le T \tag{18}$$

Here, writing $\mathcal{P}$ for partitions of [0,t], the first integral is given by

$$\int_0^t DF(X_s)\mathbf{X}_s := \lim_{|\mathcal{P}|\to 0} \sum_{[u,v]\in\mathcal{P}} (\langle DF(X_u), X_{u,v}\rangle + \langle D^2F(X_u), \mathbb{S}_{u,v}\rangle), \tag{19}$$

while the second integral is a well-defined Young integral.

Itô's formula and Föllmer

Proof (i) The first case is geometric case, $\mathbb{S} = \bar{\mathbb{S}}$, in which case the bracket is zero. Because of α-Hölder regularity of X with $\alpha > \frac{1}{3}$, we obtain

$$\begin{aligned}
F(X_T) - F(X_0) &= \sum_{[u,v]\in\mathcal{P}} F(X_v) - F(X_u) \\
&= \sum_{[u,v]\in\mathcal{P}} \langle DF\left(X_u\right), X_{u,v}\rangle + \frac{1}{2}\langle D^2F(X_u), X_{u,v}\otimes X_{u,v}\rangle \\
&+ o(|v-u|) \\
&= \sum_{[u,v]\in\mathcal{P}} \langle DF\left(X_u\right), X_{u,v}\rangle + \langle D^2F\left(X_u\right), \bar{\mathbb{S}}_{u,v}\rangle + o(|v-u|)
\end{aligned}$$

If we take the limit $|\mathcal{P}|\to 0$, then $\sum_{[u,v]\in\mathcal{P}} o(|v-u|) \to 0$.

Thus, we complete for (i)case.

(ii) The second case is non-geometric case: $\bar{\mathbb{S}}_{u,v} = \mathbb{S}_{u,v} + \frac{1}{2}[X]_{u,v}$.
Since, D^2F is Lipschitz, $D^2F(X) \in \mathcal{C}^\alpha$ and we can split-up the "bracket term" and note that:

$$\sum_{[u,v]\in\mathcal{P}} \langle D^2F(X_u), [X]_{u,v}\rangle \to \int_0^t D^2F(X_u)d[X]_u,$$

where the convergence to the Young integral follows from $[X] \in \mathcal{C}^{2\alpha}$.

We complete (ii) case, so we are done.

Remark

With regard to

$$\int_0^t DF(X_s)\,\mathbf{X}_s := \lim_{|\mathcal{P}|\to 0} \sum_{[u,v]\in\mathcal{P}} \left(\langle DF(X_u), X_{u,v}\rangle + \langle D^2F(X_u), \mathbb{S}_{u,v}\rangle\right), \tag{20}$$

in general, one can not separate the sums into two convergent sums. On the other hand, we can combine the converging sums and write :

$$F(X)_{0,t} = \lim_{|\mathcal{P}|\to 0} \sum_{[u,v]\in\mathcal{P}} \Big(\langle DF(X_u), X_{u,v}\rangle + \langle D^2F(X_u), \mathbb{S}_{u,v}\rangle \tag{21}$$

$$+ \frac{1}{2} \sum_{[u,v]\in\mathcal{P}} D^2F(X_u)\,[\mathbf{X}]_{u,v}\Big) \tag{22}$$

$$= \lim_{|\mathcal{P}|\to 0} \sum_{[u,v]\in\mathcal{P}} \left(\langle DF(X_u), X_{u,v}\rangle + \frac{1}{2}\langle D^2F(X_u), X_{u,v}\otimes X_{u,v}\rangle\right) \tag{23}$$

Itô's formula and Föllmer

Definition5.8

Let $\mathcal{P}_n$ be a sequence of patitions of [0,T] with mesh $|\mathcal{P}_n| \to 0$. We say that X : [0,T] $\to R^d$ has finite quadratic variaition in the sense of Föllmer along $|\mathcal{P}_n|$ if and only if, for evert time t$\in [0, T]$ and $1 \le i, j \le d$ the limit

$$[X^i, X^j]_t := \lim_{n\to\infty} \sum_{[u,v]\in\mathcal{P}_n} (X^i_{v\wedge t} - X^i_{u\wedge t})(X^j_{v\wedge t} - X^j_{u\wedge t}) \tag{24}$$

exists. Write $[X,X]$ for the resulting path with values in Sym($R^d \otimes R^d$), i.e. the space of symmetric $d \times d$ matrices.

Itô’s formula and Föllmer

Lemma 5.9

Assume X : [0,T] $\to R^d$ has finite quadratic variaition in the sense of Föllmer along $(\mathcal{P}_n)$. Then the map t $\mapsto [X,X]_t$ is of bounded variation on [0,T]. Assume furthermore that t $\mapsto [X,X]_t$ is continuous. Then, for any continuous G : [0,T] $\to \mathcal{L}(R^d \otimes R^d, R^e)$,

$$\lim_{n\to\infty} \sum_{[u,v]\in\mathcal{P}_n, u<t} \langle G(u), X_{u,v} \otimes X_{u,v} \rangle = \int_0^t G(u) d[X,X]_u \in R^e. \quad (25)$$

Remark

Lemma 5.9 and (21)-(23) gives the Itô-Föllmer formula:

$$F\left(X_t\right) = F\left(X_0\right) + \int_0^t DF\left(X_s\right) d\mathrm{X} + \frac{1}{2}\int_0^t D^2F(X_s)d[X,X]_t, \quad 0 \le t \le T \tag{26}$$

where first integral of RHS is given by the limit of left-point of Riemann-Stieltjes approximations

$$\lim_{n\to\infty} \sum_{[u,v]\in\mathcal{P}_n} \left\langle DF\left(X_u\right), X_{u,v}\right\rangle =: \int_0^t DF\left(X\right) d\mathrm{X} \tag{27}$$

Backward integration

Backward integration

Remark

The backward Itô's integral is defined as

$$\int_0^T f_t \overleftarrow{dB_t} = \lim_n \sum_{[s,t]\in\mathcal{P}^n} f_t B_{s,t}, \tag{28}$$

whenever this limit exists, in probability and uniformly on compact time intervals, and does not depend on the sequence of partition.
For example,

$$\int_0^t B_s \overleftarrow{dB_s} = \frac{1}{2}B_t^2 + \frac{t}{2}. \tag{29}$$

In many cases in Backward integration , integrand f may be backward adapted i.e. f_t is $\mathcal{F}_t^T$-measurable with $\mathcal{F}_s^t := \sigma(B_{u,v} : s \le u \le v \le t)$.

Backward integration

For example,

$$\int_0^t (B_t - B_s) \overleftarrow{dB_s} = \frac{1}{2} B_t^2 - \frac{t}{2}. \tag{30}$$

The backward stratonovich integral is defined as the backward Itô's integral - $\frac{1}{2}\times$ the qaudratic variation of the integrand.

Main focus and recall

The main focus in this section is to understand backward integration as rough integration.

Recall that the rough integral of (Y,Y') $\in \mathscr{D}_X^{2\alpha}$ agianst X=(X,$\mathbb{X}$) was defined by

$$\int_0^T Y d\mathbf{X} = \lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} (Y_s X_{s,t} + Y_s' \mathbb{X}_{s,t}) \tag{31}$$

where $\mathcal{P}$ are partitions of [0,T] with mesh-size $|\mathcal{P}|$.

Backward integration

Proposition 5.10(Backward representation of rough integral)

Given a rough path X=(X,$\mathbb{X}$)$\in \mathbb{C}^{\alpha}$ with $\alpha > \frac{1}{3}$ and (Y,Y') $\in \mathscr{D}_X^{2\alpha}$ we have

$$\int_0^T Yd\mathrm{X} = \lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} (Y_t X_{s,t} + Y'_t(\mathbb{X}_{s,t} - X_{s,t} \otimes X_{s,t})) \tag{32}$$

Backward integration

Proof We know that the rough integral is given as Riemann-Stieltjes limit

$$\int_0^T Y dX = \lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} (Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} + (*)_{s,t}) \tag{33}$$

whenever $(*)_{s,t} \approx 0$ in the sense that $(*)_{s,t} = O(|t-s|^{3\alpha}) = o(|t-s|)$, so that it does not contribute to the limit. But then,

$$\begin{aligned} Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} =& Y_t X_{s,t} - Y_{s,t} X_{s,t} + Y'_s \mathbb{X}_{s,t} && (34) \\ \approx& Y_t X_{s,t} - Y'_s X_{s,t} \otimes X_{s,t} + Y'_s \mathbb{X}_{s,t} && (35) \\ \approx& Y_t X_{s,t} + Y'_t (\mathbb{X}_{s,t} - X_{s,t} \otimes X_{s,t}), && (36) \end{aligned}$$

and by (33)-(36) we complete this proof.

Backward integration

Remark 5.11

Note that (32) is rewritten by

$$\int_0^T Y dX = -\lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} Y_t X_{t,s} + Y'_t \mathbb{X}_{t,s} \tag{37}$$

Note that proposition 5.10 answers the question to what extent one can replace left-point by right-point evaluation.

Note that a naively defined backward rough integral, by replacing left-point evaluation (Y_s, Y'_s) in the usual definition of the rough integral by right-point evaluation (Y_s, Y'_s), say

$$\lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} Y_t X_{s,t} + Y'_t \mathbb{X}_{s,t} \tag{38}$$

is not well defined in general.

Backward integration

Remark 5.11

By proposition, exisistence of limit (38) is equivalent to existence of

$$\lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} Y'_t X_{s,t} \otimes X_{s,t} = \lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} Y'_s X_{s,t} \otimes X_{s,t} \tag{39}$$

Backward integration

Theorem 5.12

Define the random rough paths $\mathbf{B}^{Strat} = (B, \mathbb{B}^{Strat})$ and $\mathbf{B}^{back} := (B, \mathbb{B}^{back})$ by

$$\mathbb{B}^{Strat}_{s,t} := \int_s^t B_{s,\gamma} \circ dB_\gamma = \mathbb{B}^{It\hat{o}}_{s,t} + \frac{1}{2} I(t-s) \tag{40}$$

$$\mathbb{B}^{back}_{s,t} := \int_s^t B_{s,\gamma} \overleftarrow{dB_\gamma} = \mathbb{B}^{It\hat{o}}_{s,t} + I(t-s) \tag{41}$$

Then, the following statements hold.

i) Assume that (Y(w),Y'(w)) $\in \mathscr{D}^{2\alpha}_{B(W)}$ almost surely, and Y,Y' are adpated as processes. Then, with probabilty one, for all t$\in [0.T]$,

$$\int_0^t Y d\mathbf{B}^{Strat} = \int_0^t Y_s dB_s + \frac{1}{2}\int_0^t Y_s' I ds = \int_0^t Y_s \circ dB_s \tag{42}$$

$$\int_0^t Y d\mathbf{B}^{back} = \int_0^t Y_s dB_s + \int_0^t Y_s' I ds \tag{43}$$

Backward integration

Theorem 5.12

ii) Assume that (Y(w),Y'(w)) $\in \mathscr{D}^{2\alpha}_{B(W)}$ almost surely, and Y_t, Y'_t are $\mathcal{F}^T_t$ measurable for all $t < T$. Then, with probabilty one, for all $\gamma \in [0, T]$,

$$\int_\gamma^T Y d\mathrm{B}^{Strat} = \int_\gamma^T Y_t \overleftarrow{dB_t} - \frac{1}{2}\int_\gamma^T Y'_t I dt = \int_\gamma^T Y_s \circ \overleftarrow{dB_s}, \tag{44}$$

$$\int_\gamma^T Y d\mathrm{B}^{back} = \int_\gamma^T Y_t \overleftarrow{dB_t}. \tag{45}$$

Conclusion

Summary

1. The rough integral against ($\mathbf{B} = \mathbf{B}^{It\hat{o}}$) and Itô's integral, whenever both are well-defined, coincide.

2. The rough integration against Stratonovich enhanced Brownian motion = The usual Stratonovich integration against Brownian motion under some assumptions.

3. (Itô-Föllmer formula)

$$F(X_t) = F(X_0) + \int_0^t DF(X_s)\, d\mathbf{X} + \frac{1}{2}\int_0^t D^2 F(X_s) d[X, X]_t \quad 0 \le t \le T, \tag{46}$$

4. (Backward representation of rough integral)

$$\int_0^T Y d\mathbf{X} = \lim_{|\mathcal{P}|\downarrow 0} \sum_{[s,t]\in\mathcal{P}} \left(Y_t X_{s,t} + Y'_t(\mathbb{X}_{s,t} - X_{s,t} \otimes X_{s,t})\right) \tag{47}$$

THANK YOU FOR LISTENING